

Solutions to JEE Advanced - 10 | JEE 2022 | Paper - 2

PHYSICS

SECTION-1

- 1.(B) When the piston moves downwards, the density of air trapped will increase. So, the buoyancy force due to air on the block will become slightly more and the cube will move up slightly. Similar argument holds good when the piston moves upwards.

$$2.(C) \quad I = \frac{V}{r_1 + r_2} = k_1 \theta_1 = k_2 \theta_2 \Rightarrow \frac{k_1}{k_2} = \frac{\theta_2}{\theta_1}$$

$$I'_1 = \frac{V}{r_1} = k_1 \theta'_1, \quad I'_2 = \frac{V}{r_2} = k_2 \theta'_2 \Rightarrow \frac{r_2}{r_1} = \frac{k_1 \theta'_1}{k_2 \theta'_2} = \frac{\theta_2}{\theta_1} \frac{\theta'_1}{\theta'_2}$$

$$3.(A) \quad eV_s = hv - \phi$$

$$10.4 \text{ eV} = hv - 1.7 \text{ eV}$$

$$hv = 12.1 \text{ eV} \Rightarrow n = 3 \text{ to } n = 1$$

- 4.(C) Let P be the n^{th} point of minimum

$$\text{Then, } d \sin \theta = (2n - 1) \frac{\lambda}{2}$$

$$\text{Now, the number of points of maximum between } P \text{ and } Q \text{ is } N = 2n - 1 = \frac{2d}{\lambda} \sin \theta$$

- 5.(D) Just before the collision, situation is as shown

Let the beads have velocity V_1 w.r.t. the ring and the ring has velocity V_2 .

Then, by momentum conservation $2mv_0 = (M + 2m)V_2$

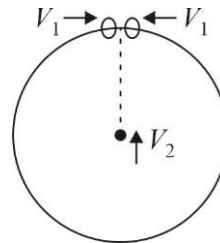
$$\Rightarrow V_2 = \frac{V_0}{2} \quad \dots\dots\dots (i)$$

By mechanical energy conservation

$$\frac{1}{2}(2m)V_0^2 = \frac{1}{2}2mv_0^2 + \frac{1}{2}2m\left(\frac{V_0}{2}\right)^2$$

Where v = velocity of beads w.r.t. ground

$$\text{So, } V = \frac{\sqrt{3}}{2} V_0$$



$$6.(C) \quad \text{Energy of satellite} = E = -\frac{GMm}{2r}$$

$$E_1 = \Delta(P.E.) = -\frac{GMm}{3R/2} - \left(-\frac{GMm}{R}\right)$$

$$E_1 = \frac{GMm}{3R}$$

$$E_2 = K.E. = \frac{1}{2} m V_{orb}^2 = \frac{1}{2} m \cdot \frac{GM}{3R/2}$$

$$E_2 = \frac{GMm}{3R} \quad \therefore \frac{E_1}{E_2} = 1:1$$

7.(C) $V = \frac{\pi}{4} D^2 \cdot h$

$$\begin{aligned} \% \text{ error} &= \frac{\Delta V}{V} \times 100\% = \pm \left(2 \frac{\Delta D}{D} + \frac{\Delta h}{h} \right) \times 100\% = \pm \left[2 \left(\frac{0.01 \text{ cm}}{4.0 \text{ cm}} \right) + \frac{0.1 \text{ cm}}{5.0 \text{ cm}} \right] \times 100\% \\ &= \pm (0.5 + 2)\% = \pm 2.5\% \end{aligned}$$

8.(BCD) Let $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$; $q\vec{V}_1 \times \vec{B} = \vec{F}$

$$\Rightarrow q\hat{i} \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) = q(-\hat{j} + \hat{k}) \quad ; \quad r = \frac{mv}{qB} = \frac{1 \times \sqrt{6}}{1 \times \sqrt{3}} = \sqrt{2} m$$

$$\Rightarrow B_y \hat{k} - B_z \hat{j} = -\hat{j} + \hat{k} \quad \therefore B_y = 1, B_z = 1$$

$$\text{Again, } q\vec{V}_2 \times \vec{B} = q(\hat{i} - \hat{k}) \Rightarrow -B_x \hat{k} + \hat{i} = \hat{i} - \hat{k}$$

$$\therefore B_x = 1 \quad \therefore \vec{B} = \hat{i} + \hat{j} + \hat{k}$$

9.(AB) The unit vector along incidence ray is

$$e = \frac{1}{\sqrt{2}} \hat{i} - \frac{1}{\sqrt{2}} \hat{j} \quad \because \mu(\hat{e} \times \hat{n}) = \mu_1(\hat{e}_1 \times \hat{n}) = \mu_2(\hat{e}_2 \times \hat{n})$$

(a) $1 \times \left(\frac{1}{\sqrt{2}} \hat{i} - \frac{1}{\sqrt{2}} \hat{j} \right) \times \hat{j} = \frac{4}{3} (x_1 \hat{i} + y_1 \hat{j}) \times \hat{j}$

$$\text{or } \frac{1}{\sqrt{2}} \hat{k} = \frac{4}{3} x_1 \hat{k} \text{ or } x_1 = \frac{3}{4\sqrt{2}} \quad \because x_1^2 + y_1^2 = 1 \text{ or } \frac{9}{32} + y_1^2 = 1$$

$$\text{or } y_1^2 = 1 - \frac{9}{32} = \frac{23}{32} \quad \therefore y_1 = \pm \sqrt{\frac{23}{32}}$$

$$\therefore \hat{e}_1 = \frac{3}{4\sqrt{2}} \hat{i} - \frac{\sqrt{23}}{4\sqrt{2}} \hat{j}$$

(b) $\because \mu(\hat{e} \times \hat{n}) = \mu_2(\hat{e}_2 \times \hat{n}) \quad \text{or} \quad \left(\frac{1}{\sqrt{2}} \hat{i} - \frac{1}{\sqrt{2}} \hat{j} \right) \times \hat{j} = \frac{3}{2} (x_2 \hat{i} + y_2 \hat{j}) \times \hat{j}$

$$\text{or } \frac{1}{\sqrt{2}} \hat{k} = \frac{3}{2} x_2 \hat{k} \quad \therefore x_2 = \frac{2}{3\sqrt{2}} \quad \because x_2^2 + y_2^2 = 1$$

$$\therefore y_2^2 = 1 - \frac{4}{18} = \frac{14}{18} \quad \therefore y_2 = \pm \frac{\sqrt{14}}{3\sqrt{2}} \quad \therefore \hat{e}_2 = x_2 \hat{i} + y_2 \hat{j} = \frac{2}{3\sqrt{2}} \hat{i} - \frac{\sqrt{14}}{3\sqrt{2}} \hat{j}$$

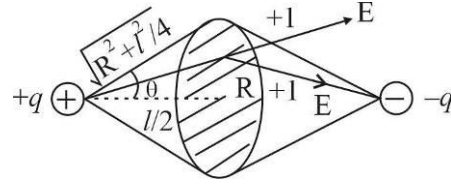
10.(ABCD)

$$\text{Flux through base of cone by each charge} = \phi = \frac{q}{2\epsilon_0}(1 - \cos \theta) = \frac{q}{2\epsilon_0} \left(1 - \frac{l}{\sqrt{4R^2 + l}} \right)$$

$$\text{Total flux} = 2\phi = \frac{q}{\epsilon_0} \left(1 - \frac{l}{\sqrt{4R^2 + l}} \right)$$

$$\text{Flux through y-z plane} = \frac{q}{\epsilon_0} (R \rightarrow \infty)$$

$$\phi = 0 \text{ as } q_{in} = 0; \quad \text{For } x^2 + y^2 + z^2 = \ell^2$$



11.(ABC)

$$\text{Acceleration : } a(t) = Pt \hat{i} + Q \hat{j}$$

$$\text{Velocity : } v(t) = v_0 + \int_0^t a(t) dt = \frac{1}{2} Pt^2 \hat{i} + Qt \hat{j}$$

$$\text{Position : } r(t) = r_0 + \int_0^t v(t) dt = \frac{1}{6} Pt^3 \hat{i} + \frac{1}{2} Qt^2 \hat{j}$$

Now, if the particle passes through the point (a, a) , then

$$\frac{1}{6} Pt^3 = a \text{ and } \frac{1}{2} Qt^2 = a$$

Both the above equations should give us the same value of t

$$\text{Therefore, } \frac{1}{6} Pt^3 = \frac{1}{2} Qt^2 \quad \dots (i)$$

Also, solving for t from both the equations separately,

$$t = \left(\frac{6a}{P} \right)^{1/3} \text{ and } t = \left(\frac{2a}{Q} \right)^{1/2}$$

At $t = T$, angle made by the velocity of the particle with the positive X-axis,

$$\theta = \tan^{-1} \left(\frac{Qt}{\frac{1}{2}Pt^2} \right) = \tan^{-1} \left(\frac{2Q}{Pt} \right) = \tan^{-1} \left(\frac{2}{3} \right) \quad (\text{using (i)})$$

12.(ACD)

$$(A) \quad \text{Time constant} = \frac{L}{R_{eq}} = \frac{2H}{8\Omega} = 0.25 \text{ sec}$$

$$\text{Just after the fuse blows } \epsilon_L + 6 = 16 \times 0.75; \quad \epsilon_L = 6V$$

$$(C) \quad i_{\infty} = \frac{E}{R_{eq}} = \frac{6V}{8\Omega} = 0.75A$$

13.(BC) Intensity of radiation reaching the spherical body,

$$I = \frac{4\pi R^2 \sigma T_0^4}{4\pi d^2} = \left(\frac{R^2}{d^2} \right) \sigma T_0^4 = \left(\frac{(10^9)^2}{(10^{11})^2} \right) (5.67 \times 10^{-8}) (6000)^4$$

$$\Rightarrow I = 7.4 \times 10^3 \text{ W/m}^2$$

Let the surface temperature of the body be T

Then, Power received from star = Power emitted

$$\Rightarrow \left[\left(\frac{R^2}{d^2} \right) \sigma T_0^4 \right] (\pi r^2) = \sigma (4\pi r^2) T^4$$

$$\Rightarrow T = \left(\frac{R}{2d} \right)^{1/2} T_0 = \left(\frac{10^9}{2 \times 10^{11}} \right)^{1/2} (6000) = 300\sqrt{2} = 423 \text{ K}$$

14.(ACD) $OA = OC = L$

$$OP = \frac{\sqrt{3}L}{2}$$

$$V_A - V_0 = \frac{\omega B L^2}{2}; \quad V_A - V_C = 0$$

$$V_C - V_0 = \frac{\omega B L^2}{2}; \quad V_P - V_0 = \frac{\omega B}{2} \left(\frac{3L^2}{4} \right) = \frac{3\omega B L^2}{8}$$

$$V_C - V_P = (V_C - V_0) - (V_P - V_0) = \frac{\omega B L^2}{2} - \frac{3\omega B L^2}{8} = \frac{\omega B L^2}{8}$$

15.(B) Use cylindrical coordinates (r, θ, z) where the z -axis is long the axis of the cable and its positive direction is the same as that of the current in the inner conductor. From $\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 i$ and the axial symmetry we have

$$B = \frac{\mu_0 i}{2\pi r}$$

From $\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\epsilon_0}$ and the axial symmetry we have

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Where λ is the charge per unit length of the inner conductor. The voltage between the conductor is

$$V = -\int_{R_2}^{R_1} \mathbf{E} \cdot d\mathbf{r}, \text{ giving}$$

$$\lambda = 2\pi\epsilon_0 V / \ln \frac{R_2}{R_1}$$

$$\text{Accordingly, } E = \frac{V}{r \ln R_2/R_1}$$

16.(D) The magnetic energy density is $w_m = \frac{B^2}{2\mu_0} = \frac{\mu_0}{2} \left(\frac{i}{2\pi r} \right)^2$. Hence the magnetic energy per unit length is

$$\frac{dW_m}{dz} = \int w_m dS = \int_{R_1}^{R_2} \frac{\mu_0}{2} \left(\frac{i}{2\pi r} \right)^2 \cdot 2\pi r dr = \frac{\mu_0 i^2}{4\pi} \ln \frac{R_2}{R_1}$$

The electric energy density is $w_e = \frac{\epsilon_0 E^2}{2} = \frac{\epsilon_0}{2} \left(\frac{V}{R \ln \frac{R_2}{R_1}} \right)^2$. Hence the electric energy per unit length is

$$\frac{dW_e}{dz} = \int_{R_1}^{R_2} w_e dS = \frac{\pi \epsilon_0 V^2}{\ln \frac{R_2}{R_1}}$$

$$\begin{aligned} \mathbf{17.(A)} \quad I_{cm} &= mR^2 + 4 \times \left[\frac{M(\sqrt{2}R)^2}{12} + M \left(\frac{R}{\sqrt{2}} \right)^2 \right] \\ &= mR^2 + \frac{2MR^2}{3} + 2MR^2 = mR^2 + \frac{8}{3}MR^2 = \left[(4)(1)^2 + \frac{8}{3}(6)(1)^2 \right] kg \, m^2 = 20 \, kg \, m^2 \end{aligned}$$

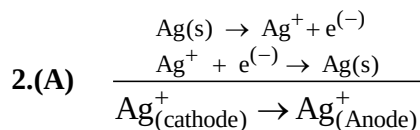
$$\begin{aligned} \mathbf{18.(B)} \quad \mu_s(\min) &\gtrsim \frac{I_{cm}}{I_{cm} + (4M + m)R^2} \tan \theta \\ &\gtrsim \frac{20}{20 + 28} \frac{1}{\sqrt{3}}; \quad \mu_s(\min) \gtrsim \frac{5}{12\sqrt{3}} \end{aligned}$$

CHEMISTRY

$$1.(C) \quad \frac{\Delta T_b}{\Delta T_f} = \frac{T_b^2 \times L_f}{T_f^2 \times L_v}$$

$$\Rightarrow \frac{0.56}{\Delta T_f} = \frac{373 \times 373 \times 80}{273 \times 273 \times 540} \Rightarrow \Delta T_f = 2$$

Freezing point of aq. solution = $273 - 2 = 271$



$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{2.303RT}{nF} \log_{10} \frac{[\text{Ag}^+]_{\text{anode}}}{[\text{Ag}^+]_{\text{cathode}}} = 0 - \frac{2.303RT}{1 \times F} \log \frac{[\text{Ag}^+]_{\text{anode}}}{[\text{Ag}^+]_{\text{cathode}}}$$

$$= 0.059 \log \frac{[\text{Ag}^+]_{\text{cathode}}}{[\text{Ag}^+]_{\text{anode}}} = 0.059 \log 10 = 0.059 \text{ Volt}$$

$$\Delta G = -nEF = \frac{-1 \times 0.059 \times 96500}{1000} = -5.7$$

$$3.(C) \quad \text{Using } \Delta_r G_{(P_2)} - \Delta_r G_{(P_1)} = (\Delta_r V)(P_2 - P_1)$$

$$P_1 = 1 \text{ bar} = 10^5 \text{ pascal}$$

$$P_2 = 1.45 \times 10^4 \text{ bar} = 1.45 \times 10^9 \text{ pascal}$$

$$\text{As } P_2 \gg P_1 \Rightarrow (P_2 - P_1) = P_2$$

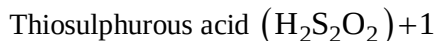
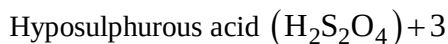
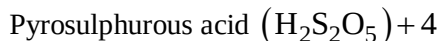
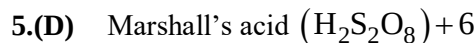
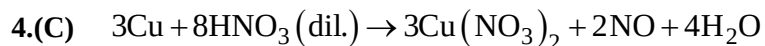
$$\text{As } P_2 \text{ is the equilibrium pressure thus } \Delta_r G_{(P_2)} = 0$$

$$\text{Also } \Delta_r G_{(P_1)} = \Delta_f G_{(D)}^\circ - \Delta_f G_{(G)}^\circ = 2.9 \text{ kJ} - 0 = 2.9 \text{ kJ}$$

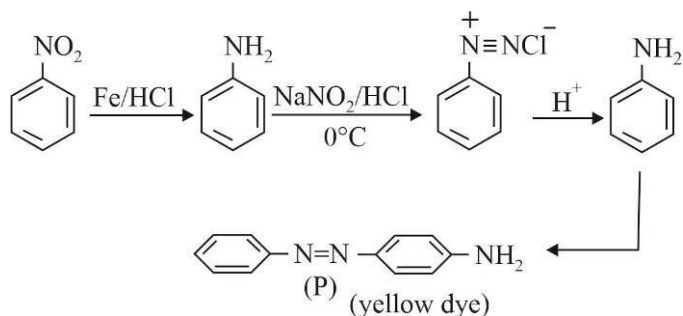
$$\Rightarrow -2.9 \times 10^3 = (\Delta_r V)(1.45 \times 10^9)$$

$$\Rightarrow \Delta_r V = -2 \times 10^{-6} \text{ m}^3 / \text{mole} = -2 \text{ ml / mole}$$

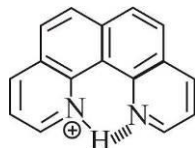
$$\Rightarrow X = 2$$



6.(B)



7.(B) Conjugate acid of R is stabilized through intramolecular H-bonding



8.(BCD)

- In adsorption process both ΔH and ΔS is negative
- Adsorption involve decrease in surface energy
- Adsorption capacity increases with increase in surface area per unit mass

9.(ABC)

- (A) As some ammonia is removed by 2nd reaction, some N_2H_4 would decompose to restore 1st equilibrium

10.(ABCD)

Using the Arrhenius equation

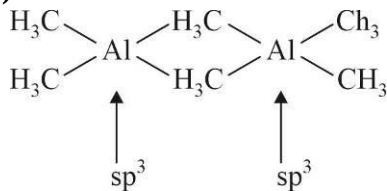
$$\Rightarrow k = A e^{-E_A/RT} = A \text{ [Boltzmann factor]}$$

$$\Rightarrow \ln \frac{k_2}{k_1} = \frac{E_A}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

It can be analysed that all the given statements are correct.

11.(AC) Both α -Halo carbonyl and benzylic position ($1^\circ/2^\circ$) are suitable for S_N2 reaction but bridgehead compound is least reactive for nucleophilic substitution reaction. This is due to unstable intermediate.

12.(AB)

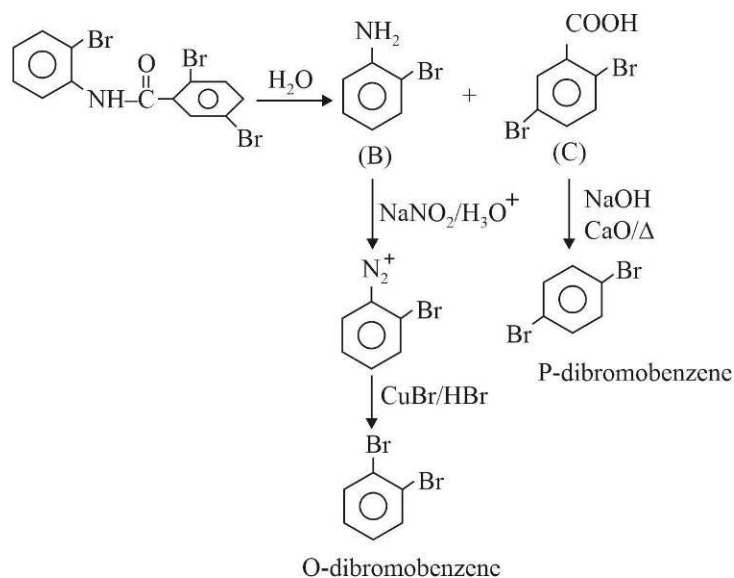


(Non planar molecule)

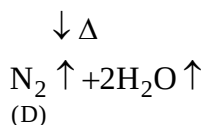
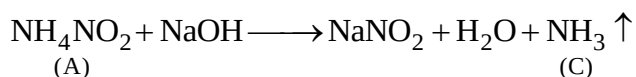
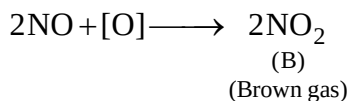
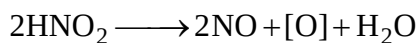
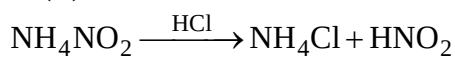
- $AlCl_3$ and Al_2Cl_6 both have vacant orbitals in central Al atom; and also C. N. of Al atoms are NOT equal to their maximum value
- Terminal B-H bonds are stronger than Bridging B-H bonds
- B_2H_6 is non planar

13.(CD) Oxides of Be / Al / Ga / Sn / Pb are amphoteric

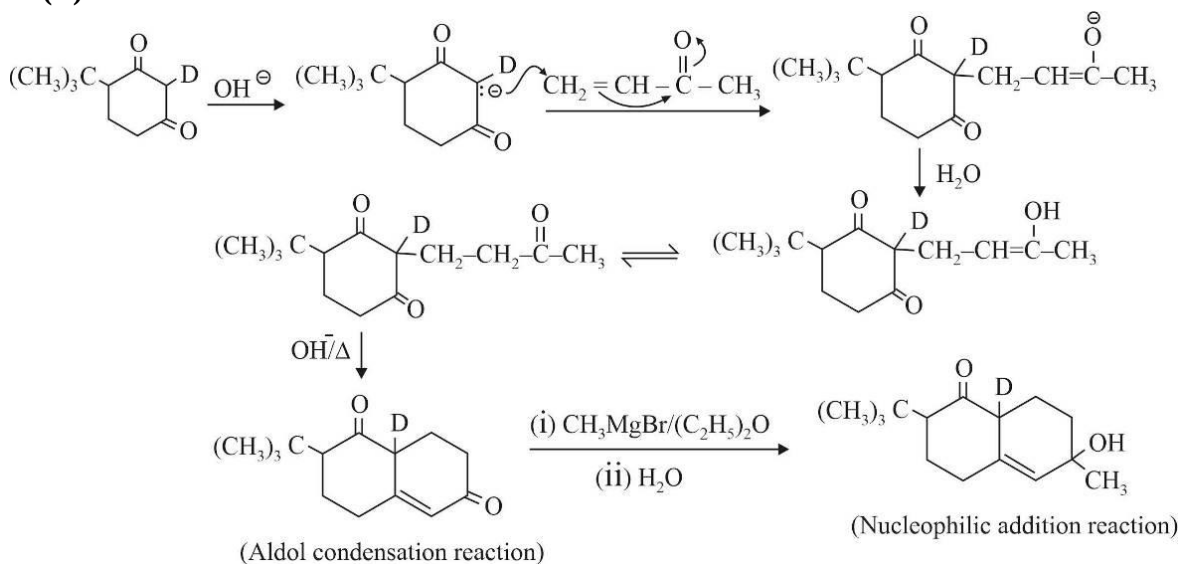
14.(ABCD)



15.(C) 16.(D)



17.(B) 18.(C)



MATHEMATICS

1.(D) $g'(x) = \frac{h'(x)}{h(x)}$

$$g''(x) = \frac{h(x)h''(x) - (h'(x))^2}{(h(x))^2}$$

$$g''(x) < 0$$

2.(B) All such lines are tangent to the circle of radius α and centre $(h, 0)$

$$\text{i.e. } y = m(x - h) + \alpha\sqrt{1 + m^2}$$

$$\frac{dy}{dx} = m$$

$$\Rightarrow y = \frac{dy}{dx}(x - h) + \alpha\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\left(y - \frac{dy}{dx}(x - h)\right)^2 = \alpha^2 \left(1 + \left(\frac{dy}{dx}\right)^2\right)$$

3.(A) $R = A^k$ and $T = B^k$

$$A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}, A^2 = \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix}, A^3 = \begin{bmatrix} 7 & 12 \\ 3 & -5 \end{bmatrix}$$

$$B^k = \begin{bmatrix} a^k & b + ab + ab^2 + \dots + ab^{k-1} \\ 0 & 1 \end{bmatrix}$$

4.(D) Let E_1 = Event of selecting male

E_2 = Event of selecting female

B = Event of colour blindness

$$P(E_1) = \frac{3}{5}; P(E_2) = \frac{2}{3}; P\left(\frac{B}{E_1}\right) = \frac{1}{2}; P\left(\frac{B}{E_2}\right) = \frac{3}{10}$$

$$\therefore P\left(\frac{E_1}{B}\right) = \frac{P(E_1) \times P\left(\frac{B}{E_1}\right)}{P(E_1)P\left(\frac{B}{E_1}\right) + P(E_2) \times P\left(\frac{B}{E_2}\right)} = \frac{\frac{3}{5} \times \frac{1}{2}}{\frac{3}{5} \times \frac{1}{2} + \frac{2}{3} \times \frac{3}{10}} = \boxed{\frac{5}{7}}$$

5.(A) $\frac{16}{(14)^4 \cdot 4} = k(1 \cdot 3 \cdot 5 \cdot 7 \cdot 9 \cdot 11 \cdot 13 \cdot 15)$

$$\Rightarrow k = \frac{2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 \cdot 12 \cdot 14 \cdot 16}{(24)^5}$$

$$\Rightarrow k = \frac{35}{27}$$

$$6.(D) \quad \left[\vec{a} \ \vec{b} \ \vec{c} \right]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = 64 \times 32$$

$$\therefore \text{Volume of parallelepiped} = 32\sqrt{2}$$

$$\text{Height of parallelepiped} = 4\sqrt{\frac{2}{3}}$$

$$\text{Volume of triangular Prism} = \frac{1}{2} [\vec{a} \ \vec{b} \ \vec{c}] = 16\sqrt{2}$$

$$7. (C) \quad \text{Let rev. line } \frac{x-1}{\ell} = \frac{y-2}{m} = \frac{z-4}{n}$$

Any point on given line $(-r+1, 3r+2, r+4)$

If $r=1 \Rightarrow$ Point is $(0, 5, 5)$

Now reflection of $(0, 5, 5)$ in plane will be $(-2, 3, 3)$

$\therefore$ dir's of line will be $-3, 1, -1$

$$\therefore \text{line is } \boxed{\frac{x-1}{-3} = \frac{y-2}{1} = \frac{z-4}{-1}}$$

$$8.(ABD) \quad f(4-x) = -f(x+4)$$

$$f(x+4) = -f(4-x)$$

$$\Rightarrow f(x) = -f(8-x)$$

$$\Rightarrow f'(x) = f'(8-x) \Rightarrow f'(10) = f'(-2)$$

$$f(x) = 2020 \quad -f(8-x) = 2020$$

$$f^{-1}(2020) = x \dots (1) \quad f(8-x) = -2020$$

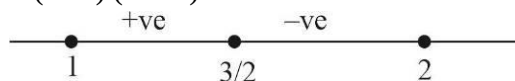
$$f^{-1}(-2020) = 8-x \dots (2)$$

$$f^{-1}(2020) + f^{-1}(-2020) = 8$$

$$\Rightarrow \int_{-2020}^4 f(x) dx = \int_{-2020}^4 f(8-x) dx$$

$$9.(BC) \quad g'(t) = f(t) - f(t-1) \forall t \in [1, 2]$$

$$= (t-2)(2t-3)$$



$g(t)$ has maxima at $t = \frac{3}{2}$ minimum value of $g(t)$ comes at either $t = 1$ or $t = 2$

$$g(1) = \frac{3}{2} \text{ and } g(2) = \frac{5}{3}$$

$$10.(BC) \int_0^x y dx = \int_0^x \left(1 + \left(\frac{dy}{dx} \right)^2 \right)^{1/2} dx$$

$$\Rightarrow \left(\frac{dy}{dx} \right)^2 = y^2 - 1 \Rightarrow y + \sqrt{y^2 - 1} = e^{\pm x}$$

$$11.(AC) \text{ Let } 7\theta = n\pi, n \in \mathbb{Z}$$

$$\Rightarrow 3\theta = n\pi - 4\theta$$

$$\tan 3\theta = \tan(n\pi - 4\theta)$$

$$\tan 3\theta = -\tan 4\theta \quad \dots(1)$$

$$\frac{3 \tan \theta - 3 \tan^3 \theta}{1 - 3 \tan^2 \theta} = - \left[\frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta} \right]$$

on simplifying, we get

$$\tan^6 \theta - 21 \tan^4 \theta + 35 \tan^2 \theta - 7 = 0$$

As $\tan \theta \neq 0$ Divide both side by $\tan^6 \theta$

$$-7 \cot^6 \theta + 35 \cot^4 \theta - 21 \cot^2 \theta + 1 = 0$$

Clearly this is cubic in $\cot^2 \theta$, having roots

$$\theta = \frac{\pi}{7}, \frac{2\pi}{7}, \frac{3\pi}{7} \quad (\text{From I})$$

$\therefore$ sum of roots

$$\cot^2 \frac{\pi}{7} + \cot^2 \frac{2\pi}{7} + \cot^2 \frac{3\pi}{7} = 5$$

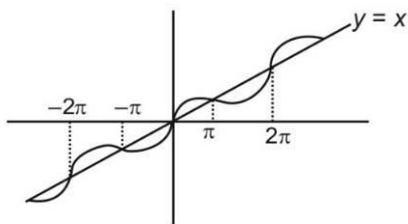
$$12.(AD) \quad f'(1+) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{|\log(1+h)| - |\log 1|}{h} = \lim_{h \rightarrow 0} \frac{|\log(1+h)|}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\log(1+h)}{h} = 1$$

$$f'(1-) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{|\log(1-h)| - |\log 1|}{-h} = \lim_{h \rightarrow 0} \frac{|\log(1-h)|}{-h} = -1.$$

$$13.(ABCD) \text{ Use the fact that } \{x\} \text{ is periodic with period } = 1 \text{ and } \int 2^x dx = \frac{2^x}{\log 2}.$$

$$14.(BCD) \quad y = x + \sin x$$



15.(A) $\vec{OR} = \vec{a} + t\vec{c}$

Given area of $\Delta ACR = 2 \Delta ABC$

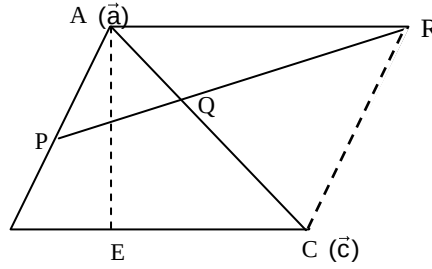
$$\text{Area of } \Delta ABC = \frac{1}{2} |\vec{c} \times \vec{a}|$$

$$\text{Area of } \Delta ACR = \frac{1}{2} |(\vec{a} + t\vec{c} - \vec{c}) \times (\vec{a} - \vec{c})|$$

$$= \frac{1}{2} |-\vec{a} \times \vec{c} + t\vec{c} \times \vec{a} - \vec{c} \times \vec{a}| = \frac{1}{2} |\vec{c} \times \vec{a}| |t|$$

$$\frac{1}{2} |\vec{c} \times \vec{a}| |t| = \frac{1}{2} |\vec{c} \times \vec{a}| \times 2$$

$$|t| = 2 \Rightarrow t = \pm 2 \Rightarrow \vec{OR} = \vec{a} + 2\vec{c}.$$



16.(B) $\vec{OP} = \frac{\vec{a}}{2}$, let $\frac{PQ}{QR} = \frac{1}{\lambda}$, $\frac{AQ}{QC} = \frac{1}{\mu}$

$$\text{position vector of Q} = \frac{\lambda \left(\frac{\vec{a}}{2} \right) + \vec{a} + 2\vec{c}}{\lambda + 1} = \frac{\mu \vec{a} + \vec{c}}{\mu + 1}$$

$$\frac{\lambda + 2}{2(\lambda + 1)} = \frac{\mu}{\mu + 1} \text{ and } \frac{2}{\lambda + 1} = \frac{1}{\mu + 1}$$

$$\text{Solving, we get } \mu = \frac{3}{2} \Rightarrow \lambda = 4$$

$$\text{p.v. of Q} = \frac{3\vec{a} + 2\vec{c}}{5}.$$

17.(A) We have $\frac{\beta}{\gamma} = \frac{1}{2} \Rightarrow \frac{\gamma}{\beta} = 2$

18.(B) We have $\alpha\beta = 1$ and $\beta\gamma = 3$; so $\frac{\gamma}{\alpha} = 3$.